HELLENIC REPUBLIC
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DEPARTMENT OF INFORMATICS & TELECOMMUNICATIONS

Artificial Intelligence II
Homework 3
Vaccine Sentiment Classifier With Bidirectional Stacked RNN Networks

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Exercise 1

All documentation for the project can be found in the three Colab notebooks.

- **HW3_AI2_VanillaRNN.ipynb:** This notebook contains the experiment for the Vaccine Sentiment Classifier With the simple RNN implementation of the pytorch libary.
- **HW3_AI2_GRURNN.ipynb:** This notebook contains the experiment for the Vaccine Sentiment Classifier With a bidirectional stacked RNN with GRU cell Neural Networks paired with GloVe embeddings.
- **HW3_AI2_LstmRNN.ipynb:** This notebook contains the experiment for the Vaccine Sentiment Classifier With a bidirectional stacked RNN with LSTM cell Neural Networks paired with GloVe embeddings.

As in the previous homework with the implementation of a Feed Forwrd neural network, my general remark is that no matter what you do in terms of hyperparameters and algorithms, the bottomline and the great difference in the final results comes from both the quality and quantity of the train and test dataset provided. Mostly the quantity and variety.

To sum up, the best results in accuracy for the test set were given from the vanilla implementation (0.53), with second best the GRU (0.52) and last was the LSTM (0.51). With the validation over the train set, the best results were coming from the LSTM (0.62), while its loss through the epochs had a nice fall. The other two, Vanilla and GRU had the same accuracy in the validation process of the test set (0.51), with minor differences in performance between the 3 classes.